



Histology of the Pancreas

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 8 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

The pancreas is best described as:

- A. a purely exocrine gland
- B. a purely endocrine gland
- C. a mucous salivary gland
- D. a mixed exocrine and endocrine gland
- E. a lymphoid gland

2.

EASY

Where are the islets of Langerhans most numerous?

- A. The head of the pancreas
- B. The tail of the pancreas
- C. The body of the pancreas
- D. The capsule
- E. The main pancreatic duct

3.

EASY

Which enzyme group digests carbohydrates?

- A. Chymotrypsin
- B. Trypsin
- C. Lipase
- D. Elastase
- E. Amylase

4.

EASY

Which islet cell secretes insulin?

- A. A (alpha) cells
- B. D (delta) cells
- C. B (beta) cells
- D. Centroacinar cells
- E. PP (F) cells

5.

MEDIUM

CASE

A pyramidal cell has basophilic cytoplasm near its base, rich in rough endoplasmic reticulum, and eosinophilic granules concentrated near its apex.

Which cell is this?

- A. Goblet cell
- B. Pancreatic acinar cell
- C. Centroacinar cell
- D. Islet beta cell
- E. Duct epithelial cell

6.

MEDIUM

What triggers the release of enzymes from acinar cells?

- A. Cholecystikinin (CCK) from duodenal enteroendocrine cells
- B. Secretin from small intestinal enteroendocrine cells
- C. Insulin from islet beta cells
- D. Gastrin from gastric G cells
- E. Somatostatin from islet delta cells

7.

MEDIUM

CASE

A cell with a condensed nucleus and clear cytoplasm sits inside the lumen of an acinus, at the point where a small duct begins.

Which cell is this?

- A. Centroacinar cell
- B. Goblet cell
- C. Intercalated duct cell
- D. Islet alpha cell
- E. Acinar cell

8.

MEDIUM

What does the centroacinar cell secrete, and what triggers it?

- A. Mucus, triggered by gastrin
- B. Digestive enzymes, triggered by CCK
- C. A watery bicarbonate-rich fluid, triggered by secretin
- D. Insulin, triggered by high blood glucose
- E. Bile, triggered by cholecystokinin

9.

MEDIUM

CASE

The smallest duct in the pancreatic system begins directly from the centroacinar cells and is lined by simple squamous to low cuboidal epithelium.

Which duct is this?

- A. Common bile duct
- B. Main pancreatic duct
- C. Intercalated duct
- D. Interlobular duct
- E. Intralobular duct

10.

MEDIUM

Which duct is formed by the joining of several intercalated ducts within one lobule?

- A. Interlobular duct
- B. Intralobular duct
- C. Main pancreatic duct
- D. Common bile duct
- E. Intercalated duct

11.

MEDIUM

A columnar duct containing occasional goblet cells and surrounded by abundant connective tissue is found between lobules, in the septa. What is it?

- A. Pancreatic acinus
- B. Centroacinar duct
- C. Intercalated duct
- D. Intralobular duct
- E. Interlobular duct

12.

MEDIUM

Where does the main pancreatic duct join the common bile duct, and what structure regulates the opening?

- A. In the pancreatic tail, regulated by local islets
- B. At the pylorus, regulated by the pyloric sphincter
- C. At the ileocecal valve, regulated by the ileocecal sphincter
- D. At the ampulla of Vater, regulated by the sphincter of Oddi
- E. At the gallbladder neck, regulated by CCK receptors

13.

MEDIUM

Which cell type secretes somatostatin, and what is its main effect in the islet?

- A. PP (F) cells; stimulate pancreatic exocrine secretion
- B. A (alpha) cells; increase glucagon secretion
- C. B (beta) cells; increase insulin secretion
- D. D (delta) cells; inhibit glucagon and insulin secretion
- E. Centroacinar cells; stimulate bicarbonate secretion

14.

MEDIUM

Which effect is attributed to pancreatic polypeptide (PP/F cells)?

- A. Increased bicarbonate secretion from centroacinar cells
- B. Increased insulin release and decreased blood glucose
- C. Decreased pancreatic exocrine secretion and gallbladder relaxation
- D. Contraction of the gallbladder and increased bile secretion
- E. Increased glucagon release and increased blood glucose

15.

HARD

CASE

A toxin selectively destroys the cells that normally respond to secretin in the pancreas.

Which structure is damaged, and which digestive process is most directly affected first?

- A. Acinar cells; neutralizing the acidic chyme
- B. Interlobular duct cells; synthesizing the enzymes
- C. Islet alpha cells; digesting dietary proteins
- D. Centroacinar cells; neutralizing the acidic chyme
- E. Islet beta cells; digesting dietary fats

16.

HARD

Which order correctly traces a drop of pancreatic secretion from an acinar cell to the duodenum?

- A. Acinar cell → intralobular duct → intercalated duct → interlobular duct → main pancreatic duct → duodenum
- B. Acinar cell → interlobular duct → intralobular duct → intercalated duct → main pancreatic duct → duodenum
- C. Acinar cell → main pancreatic duct → interlobular duct → intralobular duct → intercalated duct → duodenum
- D. Acinar cell → intercalated duct → interlobular duct → intralobular duct → main pancreatic duct → duodenum
- E. Acinar cell → intercalated duct → intralobular duct → interlobular duct → main pancreatic duct → duodenum

17.

HARD

CASE

A patient's blood glucose falls significantly between meals.

Which islet cell type would be expected to increase its secretion to compensate, and what effect would that have on blood glucose?

- A. PP cells; pancreatic polypeptide rises, raising blood glucose
- B. D (delta) cells; somatostatin secretion rises, lowering blood glucose
- C. Centroacinar cells; bicarbonate rises, raising blood glucose
- D. B (beta) cells; insulin secretion rises, raising blood glucose
- E. A (alpha) cells; glucagon secretion rises, raising blood glucose

18.

HARD

CASE

A round island of cells interrupts the deeply pink-staining tissue surrounding it, its cells separated by fine vascular channels.

Which feature distinguishes this region from the surrounding tissue, and which cell occupies most of its area?

- A. Columnar cells with goblet cells, within a duct wall
- B. Pale polygonal cells laced by capillaries; beta cells centrally, most of the area
- C. Cells with basal basophilia and apical granules
- D. Eosinophilic pyramidal cells around a lumen; acinar cells, most of the area
- E. Squamous cells, lining a small duct

19.

HARD

CASE

A pathologist finds goblet cells in the epithelium of a pancreatic duct and must decide which level of the duct system this represents.

Which explanation best accounts for the goblet cells in this duct, given the duct hierarchy?

- A. They occur in interlobular ducts, which are columnar and may contain goblet cells, unlike the smaller ducts
- B. They indicate the duct has become part of the common bile duct
- C. They occur in the main pancreatic duct, which is lined by squamous epithelium
- D. Goblet cells are a feature of every duct in the pancreatic system
- E. They occur in intercalated ducts, which are always columnar with goblet cells

20.

HARD

CASE

A disease selectively destroys the pancreatic islet cells that normally inhibit exocrine enzyme and bicarbonate secretion.

Besides digestive-enzyme secretion, which other effect would be reduced by loss of this cell type?

- A. Gallbladder relaxation — it would contract more and release more bile
- B. Mucus secretion from the duct goblet cells
- C. Bicarbonate secretion from the centroacinar cells
- D. Glucagon secretion — blood glucose would fall
- E. Insulin secretion — blood glucose would rise

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	The pancreas has an exocrine portion (acini + ducts, digestive enzymes) and an endocrine portion (islets, hormones).
2	B	Islets of Langerhans are most numerous in the tail of the pancreas.
3	E	Amylase breaks down carbohydrates; proteases and lipase handle protein and fat.
4	C	Beta cells, at the centre of the islet (~70%), secrete insulin.
5	B	Pyramidal shape, basal basophilia (RER) and apical eosinophilic zymogen granules identify the acinar cell.
6	A	CCK, produced by enteroendocrine cells of the duodenal mucosa, controls pancreatic enzyme release.
7	A	A condensed nucleus, clear cytoplasm and a position inside the acinar lumen identify the centroacinar cell.
8	C	Secretin stimulates centroacinar cells to secrete a bicarbonate-rich fluid that neutralizes acidic chyme.
9	C	The intercalated duct is the smallest, begins directly from centroacinar cells, and is lined by simple squamous or low cuboidal epithelium.
10	B	Intralobular ducts form from several intercalated ducts joining within a single lobule, lined by low columnar epithelium.
11	E	Interlobular ducts sit in the septa between lobules, are columnar and may contain goblet cells, with abundant surrounding CT.
12	D	The main pancreatic duct joins the common bile duct and opens at the ampulla of Vater, regulated by the sphincter of Oddi.
13	D	D (delta) cells secrete somatostatin, which inhibits both glucagon and insulin secretion.
14	C	PP cell secretion inhibits pancreatic exocrine secretion of enzymes and bicarbonate, and causes gallbladder relaxation with decreased bile secretion.
15	D	Centroacinar (and duct) cells provide the bicarbonate-rich fluid that neutralizes chyme; without it, pancreatic enzymes work at a suboptimal pH.
16	E	The correct order is acinar cell → intercalated → intralobular → interlobular → main pancreatic duct → duodenum.
17	E	Low blood glucose stimulates alpha cells to secrete glucagon, which raises blood glucose.
18	B	Islets are pale, polygonal, capillary-rich clusters; beta cells, centrally placed, make up about 70% of islet cells.
19	A	Only interlobular ducts are described as columnar with possible goblet cells; intercalated and intralobular ducts are not.
20	A	PP (F) cells normally relax the gallbladder and decrease bile secretion; losing them removes that inhibitory effect.

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology LG 3, Dr Hero K. Mustafa
Written from the lecture slides only – check any answer you doubt against your lecture notes.